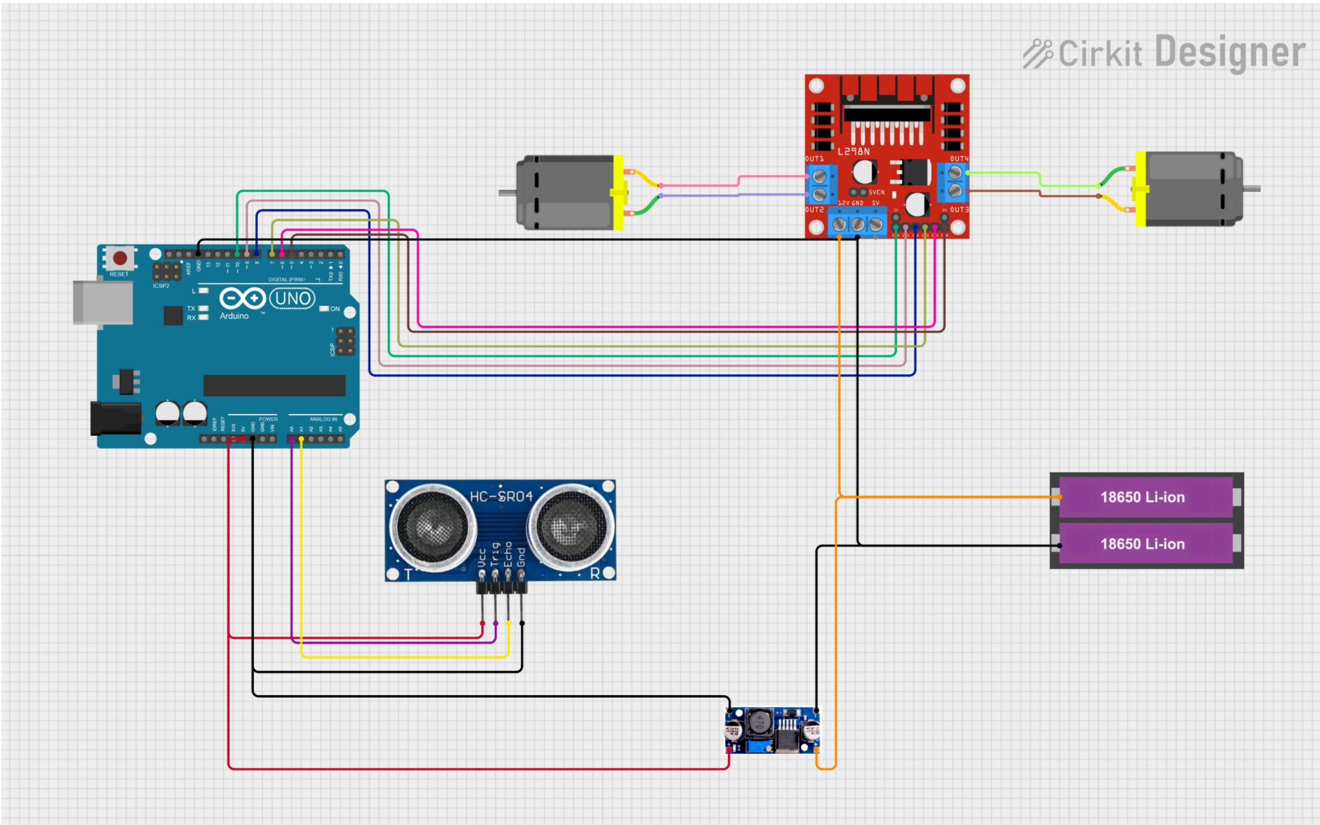
circuit diagram



Pin Connections & Wiring

Arduino Uno ↔ Ultrasonic Sensor HC-SR04

HC-SR04 Pin	Arduino Uno Pin
VCC	5V
GND	GND
TRIG	A0
ECHO	A1

Where Analog pins used as digital I/O

Arduino Uno ↔ L298N Motor Driver

L298N Pin	Arduino Uno Pin	Function
ENA	D10	Motor A speed (PWM)
IN1	D9	Motor A direction
IN2	D8	Motor A direction
IN3	D7	Motor B direction
IN4	D6	Motor B direction
ENB	D5	Motor B speed (PWM)
GND	GND	Common ground

⚠ **Remove the 5V_EN jumper on the L298N**

Motors & L298N Connections

Motor	L298N Terminal
Left Motor	OUT1 & OUT2
Right Motor	OUT3 & OUT4

Power Connections

2×18650 Li-ion Battery Pack (7.4V)

Battery	Connection
Battery +	L298N +12V (VMOTOR)
Battery –	L298N GND

Buck Converter (Step-Down Regulator)

Input Side

Buck Converter	Connection
IN+	Battery +
IN–	Battery –

Output Side (5V Regulated)

Buck Converter	Connection
OUT+	Arduino 5V
OUT–	Arduino GND

Adjust the buck converter output to exactly 5.0V before connecting to Arduino

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Common Ground (Critical)

All grounds must be connected together:

- Battery GND
- Buck converter GND
- Arduino GND
- L298N GND

code

```
/*
  Obstacle Avoidance Robot
  ------------------------
  Uses an ultrasonic sensor (HC-SR04) to detect obstacles
  and controls two DC motors via L298 motor driver.

  Author: Isra
  Date: 21 Jan, 2026
*/

#define ENA 10      // Motor A speed control
#define IN1 9       // Motor A direction
#define IN2 8
#define IN3 7       // Motor B direction
#define IN4 6
#define ENB 5       // Motor B speed control

#define TRIG_PIN A14
#define ECHO_PIN A15

int duration;
int distance;
const int SAFE_DISTANCE = 30; // cm

void setup() {
  Serial.begin(9600);

  // Motor pins
```

```
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  // Ultrasonic sensor pins
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
}

void loop() {
  distance = measureDistance();
  Serial.println(distance);

  if (distance < SAFE_DISTANCE) {
    avoidObstacle();
  } else {
    moveForward();
  }
}

// ---------------- FUNCTIONS ---------------- //

int measureDistance() {
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);
```

```
  duration = pulseIn(ECHO_PIN, HIGH);
  return duration / 58.2; // Convert to cm
}

void moveForward() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  analogWrite(ENA, 80);

  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
  analogWrite(ENB, 80);
}

void avoidObstacle() {
  // Move backward
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
  analogWrite(ENA, 80);
  analogWrite(ENB, 80);
  delay(1000);

  // Stop
  analogWrite(ENA, 0);
  analogWrite(ENB, 0);
  delay(500);

  // Turn right
```

```
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
  analogWrite(ENA, 180);
  analogWrite(ENB, 180);
  delay(800);
}
```